

-What should you orient about?

You should orient yourself in what interests you.

You should orient yourself in what the outcome is of your interests, where it leads to.

There are 10 fields in this program: AI Basics, Cybersecurity Principles & Techniques, Game Development, Intelligent Technologies, Interactive Media, Network & Cloud Basics, Process Data in Organizations, Smart Industry, Software Design & Engineering, and Open Learning.

I already have some basic experience in Software Design & Engineering. I know the basics of HTML and CSS, and just a little bit of JavaScript and Python. My main reason for joining this program is to learn how to create systems/programs/devices which can take some of our tasks and help in everyday life and this is why as a Challenge 1 I chose Project SMART HOME. But I am not closing myself only on these areas.

After exploring the other ICT topics with my group, I'm curious and motivated to learn about things like AI, Cybersecurity Principles & Techniques Game Development, Intelligent Technologies, Network & Cloud Basics, Process Data in Organizations, and Smart Industry. I'd like to explore all of them, because that will help me figure out which fields I like best and which one I'd like to focus on in the future, and it will also give me the opportunity to grow in many areas.

Below is our interpretation of the ICT fields, which we created together with my group.

AI BASICS

How AI reacts based on your inputs, what is the backend behind AI, What is used to create an AI, What information does it process. What can be the benefits and uses of AI.

Profession: AI Engineer, Prompt Engineer.

Giving negative feedback on AI's mistake, transcribing the AI's decisions, trying to understand what the AI is doing and why.

CYBER SECURITY , PRINCIPLES & TECHNIQUES

How you should handle your own hardware, what type of attacks exist in cybersecurity, improving your own cybersecurity.

Profession: PenTester, Cyber Security Engineer

They try to find vulnerabilities in a system or application and then report later what happens and why it does. Otherwise they are monitoring the system and or application for detections about threat actors.

GAME DEVELOPMENT

Creating and visualizing a game.

Profession: Game Developer

Creates and visualizes a game.

INTELLIGENT TECHNOLOGIES

Intelligent Technologies is the use of software, products and or robotics that are smart, they analyze, predict and react to data. They can “think” in an algorithm using inputs.

Profession: Robotics Engineer

Design, code and fix the products, software and or robotics that they manage.

INTERACTIVE MEDIA

You design websites and focus more on the user interface and user experience.

You visualize the websites and or other interactive media format.

Profession: UI/UX Engineer

Improves and designs UI/UX, processes user experience feedback and integrates that into the already existing design.

NETWORK CLOUD & BASICS

Creating, designing and managing the infrastructure of a company or entities servers.

Using various platforms.

Profession Cloud Network Engineer

They could be onsite or off-site managing the cloud, making sure that nothing goes wrong and if it does that it's fixed as fast as possible.

OPEN LEARNING

You decide what you learn from the other topics, you are responsible for everything.

Profession: Depends on what you chose to do in your open learning.

Anything.

PROCESS DATA IN ORGANIZATIONS

You analyze the company workflow, and visualize it. You try to predict and or make chooses that help the company / entity you work for profit more.

Profession: Data Analyst

Analyzes the data of the company's workflow and come up with solutions for potential problems.

SMART INDUSTRY

Improve the quality of your product and or the performance of your company using data, you are trying to improve the workflow without losing quality. You can use various methods of processing your data.

Profession: Smart Industry Engineer

You process the workflow of your company/product to find improvements/optimizations.

SOFTWARE DESIGN & ENGINEERING

You design,plan and create the product that you visualize. You are the one maintaining and testing the software and you are also the one updating software.

Profession: Software Engineer

Writing code, design systems, you test the system and you design the software.

How do you think you can provide evidence for this learning outcome in your portfolio?

You can explain your project in a way that showcases that you created it, and you learned and thought about it. You can describe your process, showcasing your inspiration, explaining why.

UPDATES

BLOCK 1

29.09.2026

UPDATE AFTER CHALLENGE 1

I've just finished my first project, and I can confidently say that it allowed me to explore a little bit deeper some of the topics that I used in it. I was building Smart Home system using Intelligent Technologies partly because I was working with sensors and components (I wrote partly because I planned more functions than I finally made, it's just because I focused too much on preparing documentation than on physically building it but I already had conversation and feedback about this with my semestral coach) and also Software Design & Engineering to write code in C++.

After understanding enough logic to build my function (which wasn't easy) this programming language was enjoyable to use. The part that I enjoyed the most in this project was working with components, sensors and controlling them, building something smart and controlling it. Next aspect that I like in this project is how devices actually work and communicate with each other. I used just a few components and sensors but knowledge about them for sure will stay in my mind. Honestly I feel a little bit upset that I didn't fully finish my project. I could use more fields for instance:

- **Network & Cloud Basics** (for application to control system connected by Bluetooth/Wi-Fi)
- **Process Data in Organizations** (to collect data from sensors and analyze them)
- **Cybersecurity** (I planned an RFID lock and alarm system)

In the future, I will do more small projects in my free time, because **this project showed me that I enjoy Intelligent Technologies combined with Software Design & Engineering.**

BLOCK 2

29.09.2026

UPDATE BEFORE CHALLENGE 2

As my next project, I chose Security Playground because it includes topics like Cyber Security Principles & Techniques and Network & Cloud Basics, which I **have been curious about since the start of the semester.**

I compared it with all the other options. Future Factory looked interesting, but it felt similar to Smart Home, since it also involved sensors and hardware, just in a virtual environment. (A)I Robot also looked interesting, but I was worried it would reuse the same topics as Smart Home, such as Smart Industry, Intelligent Technologies, and Software Design & Engineering, instead of teaching me

something new. Sports Community felt comfortable, since I already have experience building websites, but I get bored quickly with web development and do not see myself doing it long-term. Greenhouse Management felt almost identical to Smart Home.

The Passion Challenge, where I considered building a simple game, started to interest me after seeing my friends do it for Challenge 1, but Security Playground was the option that felt genuinely new to me and matched my curiosity best. I want to learn how ethical hacking works, how networks and machines actually communicate with each other, and later how to prevent different types of attacks.

01.10.2026

NETWORKING & CLOUD BASICS WORKSHOP

Today I had a workshop on Networking & Cloud Basics. We set up GNS3 and had a short talk about the topic. This was useful, because my next project, Security Playground, is based on networking, so I will use GNS3 directly for it.